

CAT2 Key

Degree & Branch	BE CSE				Semester	V
Subject Code & Name	UCS 2501 Computer Networks					
Academic Year	2023-2024 ODD	Batch	2021-2025	Date	02.11.2023	FN
Time: 08:10 - 09:40 a.m (90 Minutes)	Answer All Questions				Maximum: 50 Marks	

1. UDP efficiency = Useful bytes / Total bytes = 16/24 = 66.67%

2. **FTP Data- 20 , control – 21 port numbers. – (1mark)**

Usage:

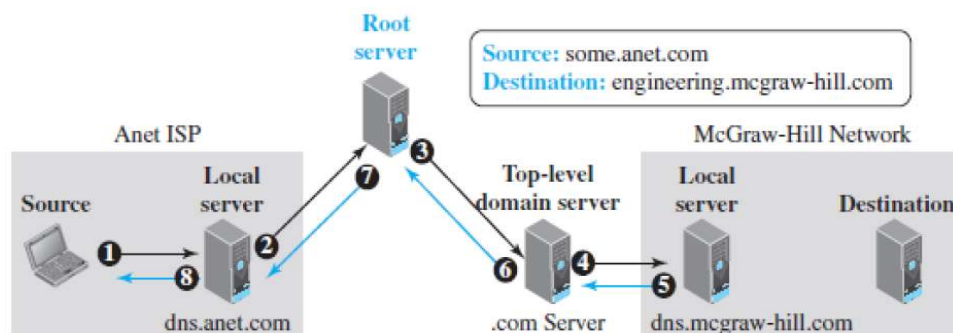
FTP is used for transferring a file from one host to another. The two connections in FTP have different lifetimes. The control connection remains connected during the entire interactive FTP session. The data connection is opened and then closed for each file transfer activity. It opens each time commands that involve transferring files are used, and it closes when the file is transferred. – **1 mark**

3. To prevent deadlock, when the persistence timer goes off, the sending TCP sends a special segment called a *probe*. The threshold value of the persistence timer is 60secs.

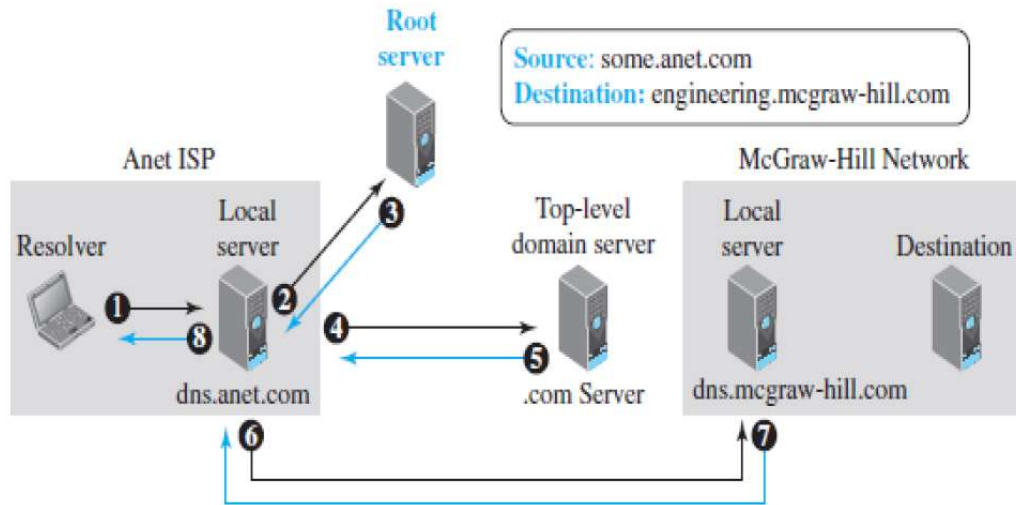
4. **Rule No 6** is applied. If a duplicate segment arrives, the receiver discards the segment, but immediately sends an acknowledgment indicating the next in-order segment expected. This solves some problems when an **ACK segment itself is lost**.

5. DNS

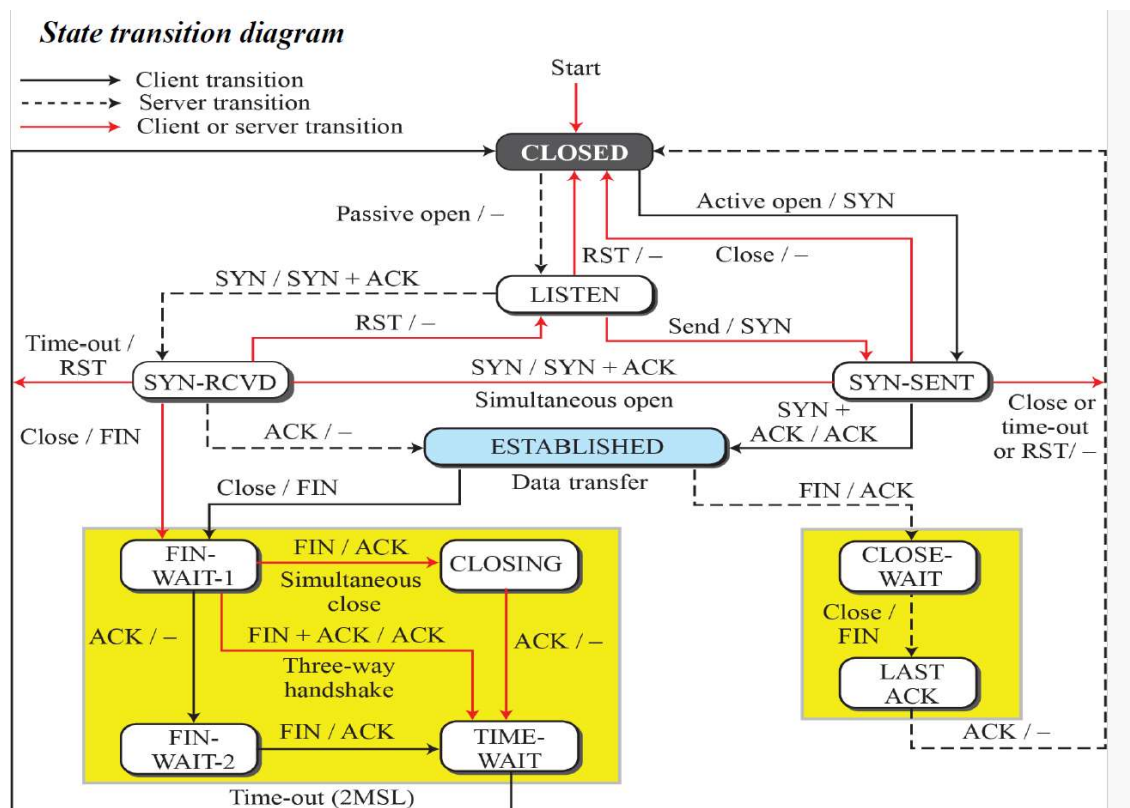
Recursive resolution



Iterative resolution



6.TCP State transition diagram



7. DV Vs LS

	DV	LS
Information Exchange	It shares directing data with their quick neighbors as it were.	It trades total organized topology data with all other routers within the organization.
Routing Table	It intermittently upgrades its directing tables and shares them with neighbors.	It autonomously calculates the most limited way to each goal based on the total organize topology.
Knowledge of Network	Network Each router has restricted information about the organization past its quick neighbors	Each router keeps up a comprehensive see of the whole network's topology.
Convergence Time	Joining is slower, particularly in bigger systems, due to incremental upgrades and restricted data sharing.	Merging is quicker as routers have total and up-to-date information on the network's topology.
Resource Usage	It requires less memory and handling control since switches as it were ought to store data around their neighbours.	It requires more memory and preparing control to store and prepare the total to organize topology data.
Protocol	RIP	OSPF

8. Address allocation:

Answer key:

Allocate from largest subblock to smallest subblock:

First department: $2^9=512, n=32-9=23$

32 teams , each with 512 address

First customer: 203.50.0.0/23 to 203.50.0.255/23, 203.50.1.0/23 to 203.50.1.255/23

Second customer: : 203.50.2.0/23 to 203.50.2.255/23, 203.50.3.0/23 to 203.50.3.255/23

Third customer: 203.50.4.0/23 to 203.50.4.255/23, 203.50.5.0/23 to 203.50.5.255/23

....

Last customer: 203.50.62.0/23 to 203.50.62.255/23, 203.50.63.0/23 to 203.50.63.255/23

Second department:

$2^8=256, n=32-8=24$

64 teams , each with 256 address

First customer: 203.50.64.0/24 to 203.50.64.255/24
 Second customer: : 203.50.65.0/24 to 203.50.65.255/24
 Third customer: 203.50.66.0/24 to 203.50.66.255/24

Last customer: 203.50.127.0/24 to 203.50.127.255/24

Third department:

$2^7=128$. $N=32-7=25$

128 teams , each with 128 address

First customer: 203.50.128.0/25 to 203.50.128.127/25

Second customer: : 203.50.128.128/25 to 203.50.128.255/25

Third customer: 203.50.129.0/25 to 203.50.129.127/25

...

Last customer: 203.50.191.128/25 to 203.50.191.255/25

Total available addresses in the /16 block = 65,536 addresses

Total addresses used = 16,384 (first department) + 16,384 (second department) + 16,384 (third department) = 49,152 addresses

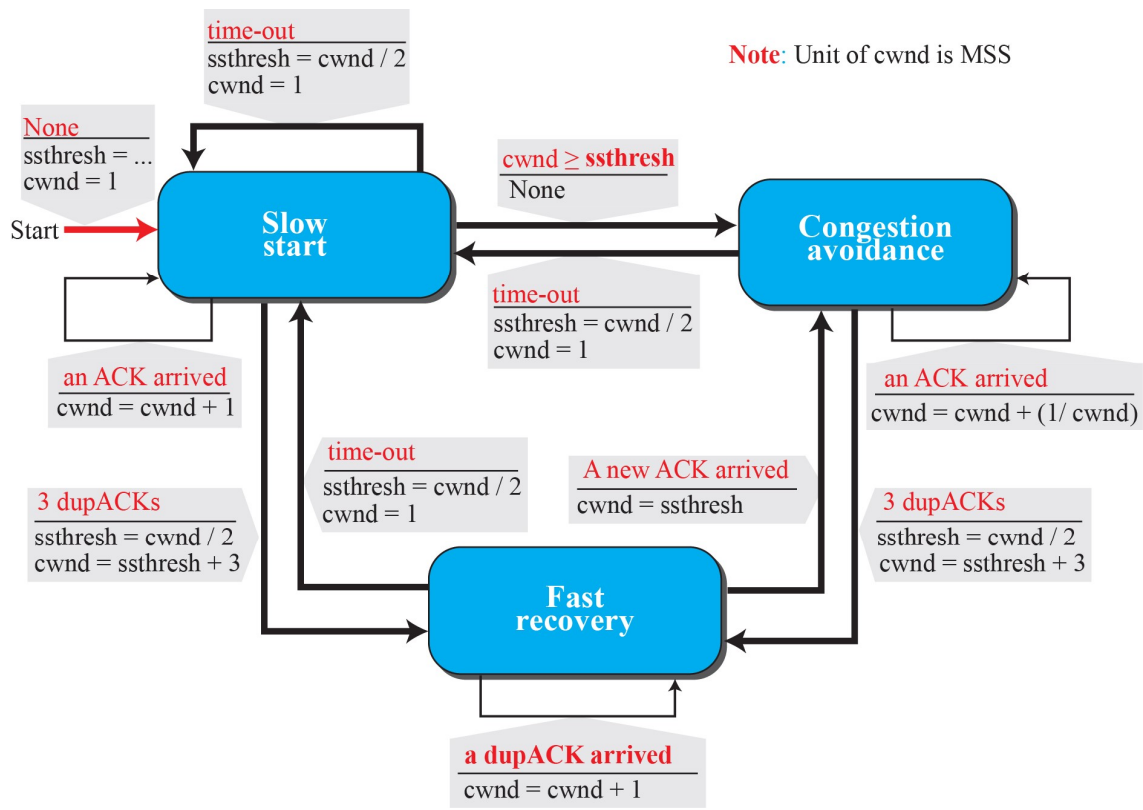
Addresses still available = 65,536 - 49,152 = 16,384 addresses

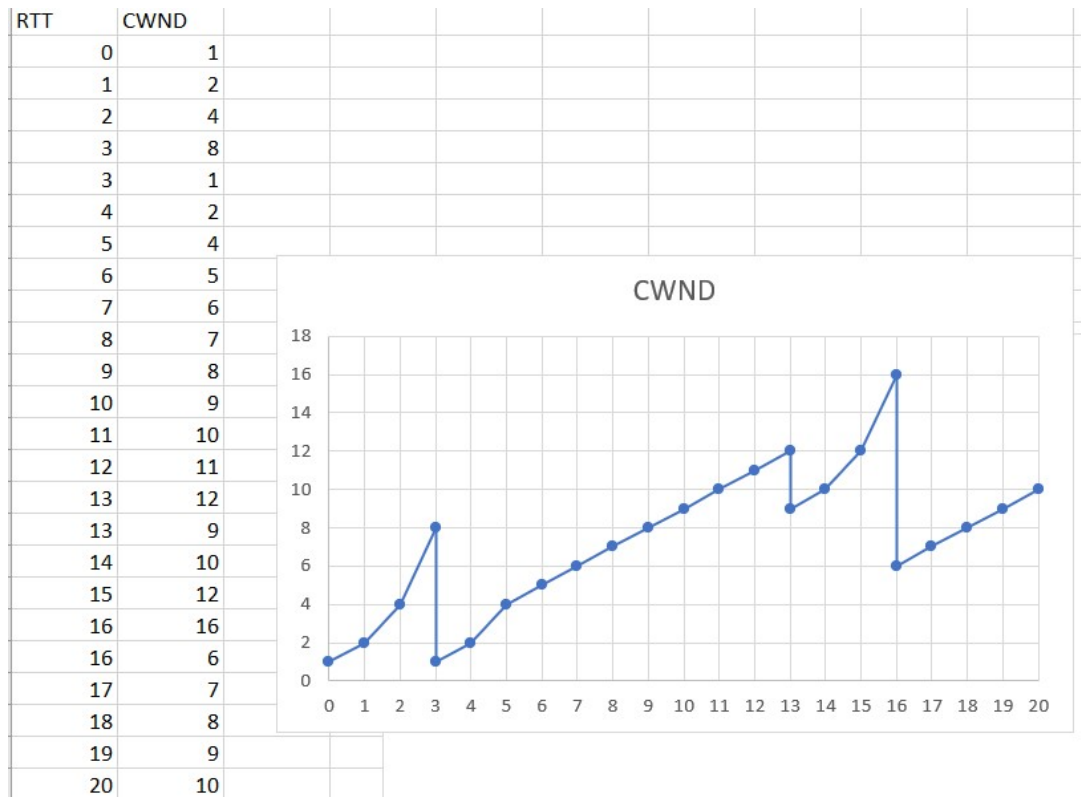
9. DV:

Node E:

Node E	From D (5)	New - E	from B (4)	New E from F (4)	from F (4)
A 2	A 2	8 A	2	6 2	2 2 B
B 4	B 2	4	0	4 2	4 4
C 2	C 2	2	5	9 B 4	6 F
D 5	D 0	5	2	5 2	5
E 0	E 5	0	4	0 2	0
F 2	F 2	2	2	2 0	2 2
G 2	G 2	2	2	2 1	2 F

10. TCP Reno:





11. Goback N, SR

